

This is a walkthrough of the Angular environment setup. We will install the Angular CLI and Material, the Angular UI framework. We will add a simple row of dummy buttons, and a basic list of TaskManager users.

- The NVM for Windows utility makes it very convenient to manage npm and Node.js, which get version updates very frequently. You can download NVM from <https://github.com/coreybutler/nvm-windows>. The commands used below are being executed after installing NVM and using `nvm run latest` (which is installing Node.js v25.3.0 as of 1/14/26). The current version of Angular (v21.0.5) is not supported by Angular, but we should still be able to execute our code with it. Our textbook is using Angular v13, so some things will change.

```
added 312 packages in 23s
npm notice
npm notice New minor version of npm available! 11.6.2 -> 11.7.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.7.0
npm notice To update run: npm install -g npm@11.7.0
npm notice
```

- Node.js version v25.3.0 detected.
Odd numbered Node.js versions will not enter LTS status and should not be used for production. For more information, please see <https://nodejs.org/en/about/previous-releases/>.

```
Angular CLI      : 21.0.5
Node.js         : 25.3.0 (Unsupported)
Package Manager : npm 11.6.2
Operating System : win32 x64
Warning: The current version of Node (25.3.0) is not supported by
Angular.
```

- Clone the project repo and navigate to the project folder. The TaskManager folder is the solution from the final project from CEN3024C, but with the GH actions disabled.

cd CEN4025C-M1Project

dir

Volume in drive C has no label.

Volume Serial Number is 84EF-A13F

Directory of C:\Users\dasingle\Downloads\CEN4025C-M1Project

```
01/13/2026 09:06 AM <DIR>      .
01/13/2026 09:05 AM <DIR>      ..
01/13/2026 09:06 AM <DIR>      .github
01/13/2026 09:06 AM      4,278,910 CEN4025C-Module1Project.pdf
01/13/2026 09:06 AM          22 README.md
01/13/2026 09:06 AM <DIR>      TaskManager
                2 File(s)  4,278,932 bytes
                4 Dir(s) 103,711,088,640 bytes free
```

- Use the **ng new** command to create the front end project. Press <enter> as many times as necessary to accept all of the defaults.

ng new TaskManagerFrontEnd

ng new TaskManagerFrontEnd

Node.js version v25.3.0 detected.

Odd numbered Node.js versions will not enter LTS status and should not be used for production.

For more information, please see <https://nodejs.org/en/about/previous-releases/>.

Would you like to share pseudonymous usage data about this project with the Angular Team at Google under Google's Privacy Policy at <https://policies.google.com/privacy>. For more details and how to change this setting, see <https://angular.dev/cli/analytics>.

No

Global setting: disabled

Local setting: No local workspace configuration file.

Effective status: disabled

√ Which stylesheet system would you like to use? CSS

[<https://developer.mozilla.org/docs/Web/CSS>]

√ Do you want to enable Server-Side Rendering (SSR) and Static Site Generation (SSG/Prerendering)? No

√ Which AI tools do you want to configure with Angular best practices?

<https://angular.dev/ai/develop-with-ai> None

CREATE TaskManagerFrontEnd/angular.json (2015 bytes)

CREATE TaskManagerFrontEnd/package.json (969 bytes)

CREATE TaskManagerFrontEnd/README.md (1531 bytes)

```
CREATE TaskManagerFrontEnd/tsconfig.json (990 bytes)
CREATE TaskManagerFrontEnd/.editorconfig (331 bytes)
CREATE TaskManagerFrontEnd/.gitignore (647 bytes)
CREATE TaskManagerFrontEnd/tsconfig.app.json (444 bytes)
CREATE TaskManagerFrontEnd/tsconfig.spec.json (456 bytes)
CREATE TaskManagerFrontEnd/.vscode/extensions.json (134 bytes)
CREATE TaskManagerFrontEnd/.vscode/launch.json (490 bytes)
CREATE TaskManagerFrontEnd/.vscode/tasks.json (1020 bytes)
CREATE TaskManagerFrontEnd/src/main.ts (228 bytes)
CREATE TaskManagerFrontEnd/src/index.html (318 bytes)
CREATE TaskManagerFrontEnd/src/styles.css (81 bytes)
CREATE TaskManagerFrontEnd/src/app/app.spec.ts (709 bytes)
CREATE TaskManagerFrontEnd/src/app/app.ts (313 bytes)
CREATE TaskManagerFrontEnd/src/app/app.css (0 bytes)
CREATE TaskManagerFrontEnd/src/app/app.html (20446 bytes)
CREATE TaskManagerFrontEnd/src/app/app.config.ts (324 bytes)
CREATE TaskManagerFrontEnd/src/app/app.routes.ts (80 bytes)
CREATE TaskManagerFrontEnd/public/favicon.ico (15086 bytes)
√ Packages installed successfully.
  Directory is already under version control. Skipping initialization of git.
```

- Move to the project folder and start VSCode

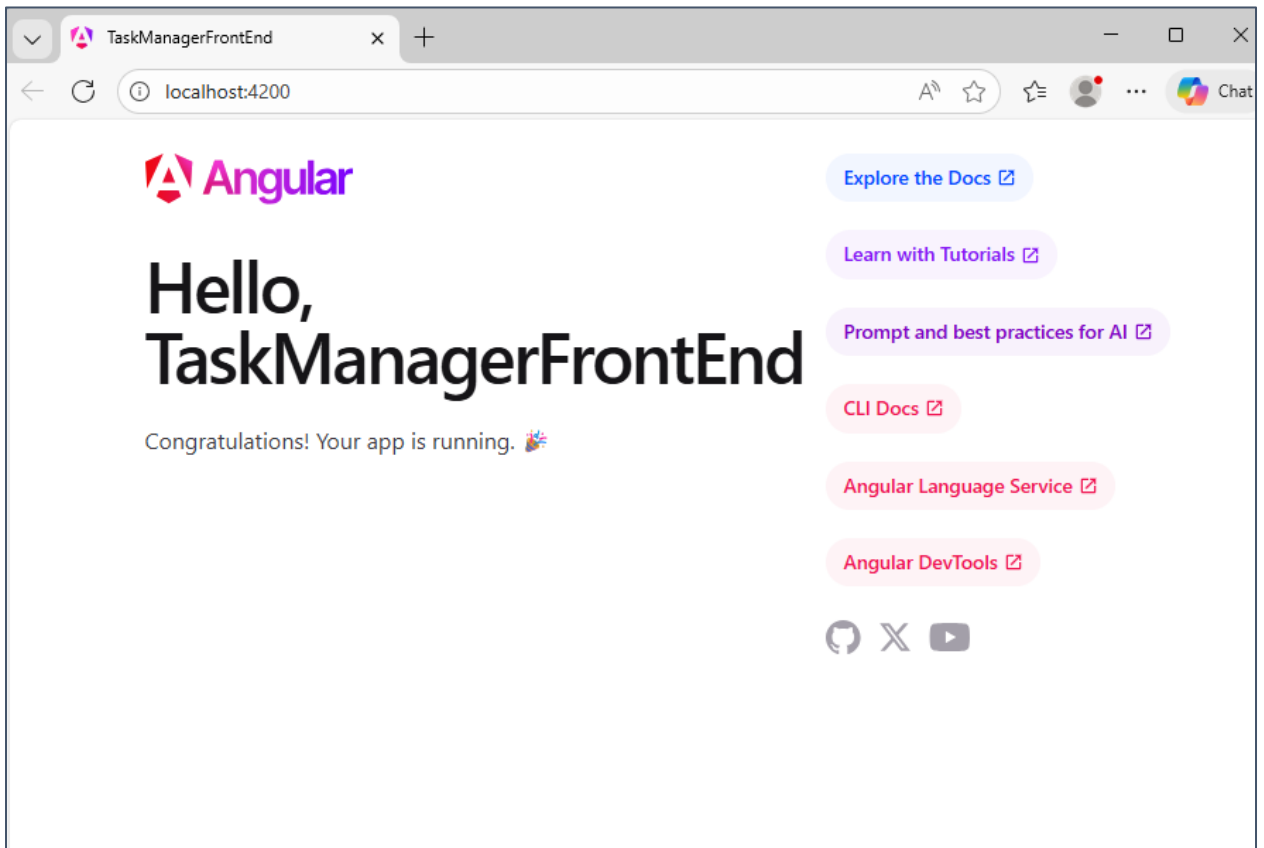
```
cd TaskManagerFrontend
code .
```

- Start a terminal, then build and run the server:

```
ng serve
```

PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL
			<pre>polyfills.js polyfills 85.81 kB main.js main 21.99 kB styles.css styles 96 bytes Initial total 107.89 kB Application bundle generation complete. [1.574 seconds] Watch mode enabled. Watching for file changes... → Local: http://localhost:4200/ → press h + enter to show help</pre>

- Use the link provided to test the install



- Ctrl-C to break out of the server in the VS Code Terminal. You can also take advantage of the watch feature; changes made in the project are automatically pushed ("hot reloaded") out to the running server without having to stop/rebuild/start.
- Install the Materials UI framework in your project and select the various associated options:

```
ng add @angular/material
```

```
Node.js version v25.3.0 detected.
```

```
Odd numbered Node.js versions will not enter LTS status and should not  
be used for production. For more information, please see  
https://nodejs.org/en/about/previous-releases/.
```

- ✓ Determining Package Manager
 - › Using package manager: npm
- ✓ Searching for compatible package version
 - › Found compatible package version: 21.1.0.
- ✓ Loading package information from registry
- ✓ Confirming installation
- ✓ Installing package

✓ Select a pair of starter prebuilt color palettes for your Angular Material theme Azure/Blue [Preview:
<https://material.angular.dev?theme=azure-blue>]

UPDATE package.json (1036 bytes)

✓ Packages installed successfully.

CREATE src/material-theme.scss (1179 bytes)

UPDATE angular.json (2057 bytes)

UPDATE src/index.html (651 bytes)

- Note the changes to package.json when you add components; this allows an environment to be reproduced by running npm install against the file, similar to the pom.xml file in our Java projects.
- Select an example button bank from <https://material.angular.io/components/button/examples> and copy the code to app.html to display it. I added my button code to app.component.html, just below the logo:

```
<div class="example-label">Filled Button Examples</div>
<div class="example-button-row">
  <button matButton="filled">Basic</button>
  <button matButton="filled" disabled>Disabled</button>
  <a matButton="filled" href="https://www.google.com/"
target="_blank">Link</a>
</div>
```

```
<h1>Hello, {{ title }}</h1>
```

```
<p>Congratulations! Your app is running. 🎉</p>
```

